

EEE4114F Final Exam

June 2022

Duration: 3 hours

Total: 100 marks

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 6 questions totalling 55 marks. You must answer all of them.
- **Part B** has 4 questions totalling 45 marks. You must answer all of them.
- Part A and Part B must be answered in separate exam scripts.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 3 hours.

Part A:	Question:	1	2	3	4	5	6	Total
	Points:	10	5	10	10	10	10	55
	Score:							

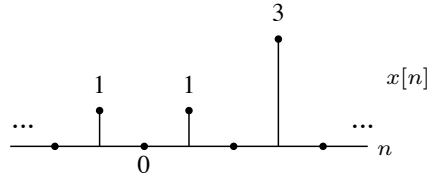
Part B:	Question:	7	8	9	10	Total
	Points:	10	10	12	13	45
	Score:					

Part A: Digital Signal Processing

Parts A and B must be answered in separate exam booklets.

Question 1 [10 marks]

If $x[n]$ is the signal

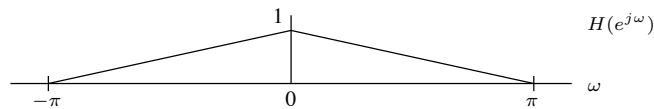


then plot the signals

- $y_1[n] = 1 - 2x[n]$
- $y_2[n] = -2x[2n]$
- $y_3[n] = -2x[-n] + 2$
- $y_4[n] = x[n]\delta[n - 1]$
- $y_5[n] = x[-n]u[2 + n]$

Question 2 [5 marks]

The frequency response $H(e^{j\omega})$ of a discrete-time LTI system is as shown below:



$H(e^{j\omega})$ is real valued so the phase is zero. Find the output $y[n]$ when the input $x[n]$ is

$$x[n] = 1 + \cos(0.3\pi n).$$

Express your solution in a form that does not contain any complex quantities.

Question 3 [10 marks]

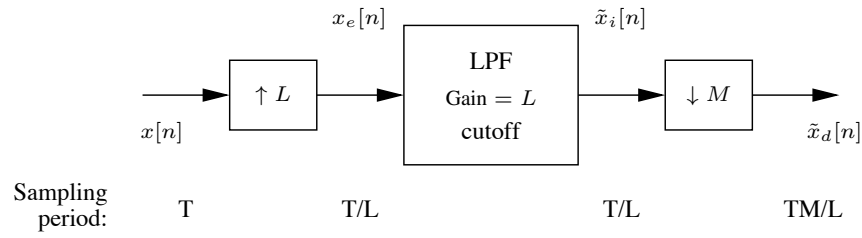
Consider the causal system whose input $x(t)$ and output $y(t)$ are related by the difference equation

$$y[n] = 5/6y[n - 1] - 1/6y[n - 2] + x[n] - 1/2x[n - 1].$$

Find the output when the input is a unit step.

Question 4 [10 marks]

Consider the system



where the LPF is assumed to be ideal with a cutoff at $\min(\pi/L, \pi/M)$. Suppose the input is

$$x[n] = \sin(4\pi n/5)/(\pi n),$$

with $L = 6$ and $M = 5$.

- Sketch the spectrum $X(e^{j\omega})$ of $x[n]$.
- Sketch the spectrum $X_e(e^{j\omega})$ of $x_e[n]$.
- Sketch the spectrum $\tilde{X}_i(e^{j\omega})$ of $\tilde{x}_i[n]$.
- Sketch the spectrum $\tilde{X}_d(e^{j\omega})$ of $\tilde{x}_d[n]$.
- What is the overall change in sampling rate?

Question 5 [10 marks]

A real-valued discrete time signal $x[n]$ is of length $N = 6$. We are given several values of its discrete Fourier transform (DFT):

$$X[0] = 10, \quad X[1] = 1 + 2j, \quad X[2] = 2 - j, \quad X[3] = 2.$$

- Determine the missing values of $X[k]$ for $k = 4$ and $k = 5$.
- We are given another length-6 discrete time signal $y[n]$ whose DFT, $Y[k]$, is related to $X[k]$ as follows:

$$Y[k] = X[k](1 + e^{-j\pi k}).$$

Compute $Y[k]$ for $k = 0, 1, \dots, 5$. (If you are not sure about your answers in part (a) you may assume that $X[4] = 6 - 5j$ and $X[5] = -7 + 2j$.)

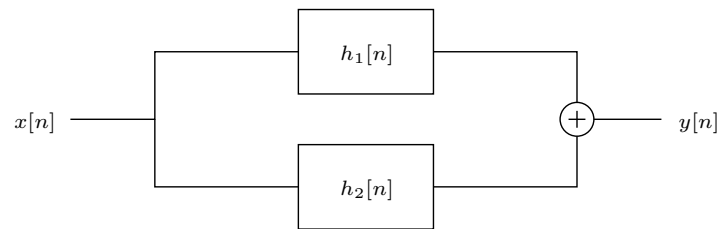
- Consider the DFT time shift property

$$x[((n - m))_N] \iff X[k]e^{-j2\pi km/N},$$

where $x[((n - m))_N]$ denotes time indices that are cyclical with modulo N . Use the time shift property to express $y[n]$ in terms of $x[n]$. (We expect a response such as $y[n] = 2x[n] - 17$ or $y[n] = x[((n - 1))_N] - x[((n - 3))_N]$.)

Question 6 [10 marks]

Consider the parallel combination of two LTI systems below:



You are told that

$$h_1[n] = u[n] - 2u[n-1] + u[n-2],$$

and you observe that the step response of the overall system is

$$s[n] = 2r[n] - 3r[n-1] + r[n-2]$$

with $r[n] = nu[n]$.

- Find $H_1(z)$.
- Find $S(z)$.
- Show that

$$H_2(z) = \frac{-1 + 4z^{-1} - 4z^{-2} + z^{-3}}{1 - z^{-1}}$$

with ROC $|z| > 1$.

- Find and sketch $h_2[n]$.

END OF PART A

Part B: Machine Learning

Parts A and B must be answered in separate exam booklets.

Question 7: Model evaluation [10 marks]

Suppose you train a model with gradient descent optimization multiple times using cross-validation. The results of the training and validation loss are plotted in Figure 1 for three different splits of the data.

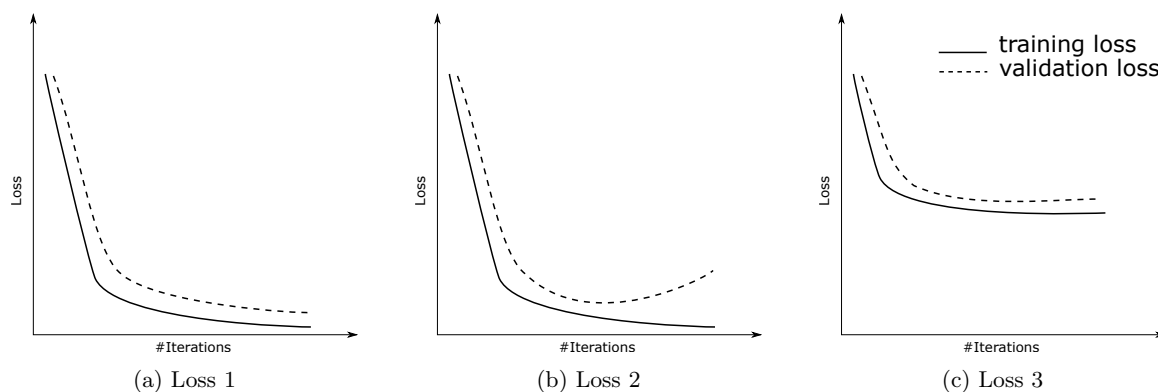


Figure 1: Training and validation loss vs number of training iterations. The training loss is depicted with a solid line, and the validation loss with a dotted line.

- Does this model have high or low variance? Explain your answer. (2)
- Which loss curve shows signs that the model is overfitting to the training data? Explain your answer. (2)
- Explain why it can be problematic to evaluate a model using the parameters as they are after the final training iteration. (2)
- Which loss curve would indicate a model with high bias compared to the others? Explain your answer. (2)
- Would you expect this model to have high or low capacity? Explain your answer. (2)

Question 8: k-Nearest Neighbours [10 marks]

You are given the task of classifying fruits into two classes, bananas and mangoes, based on two features x_1 and x_2 . You decide to use a k-Nearest Neighbour (kNN) algorithm to classify the fruits using Euclidean distance as the distance metric. Figure 2 shows the training samples as well as two test samples. You can consider mangoes to be class '0' and bananas to be class '1' to make it simpler to write. Answer the following questions.

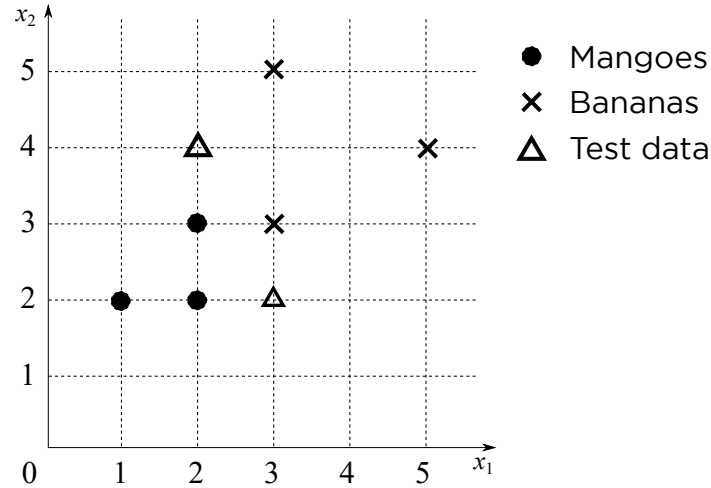


Figure 2: A plot of training and test examples with respect to features x_1 and x_2 .

- Is the variable, k , in a kNN algorithm a model parameter or a hyperparameter? Explain your answer. (2)
- If $k = 3$, how would the kNN algorithm classify each test point, T_1 and T_2 . Show all working out and steps that the kNN would take. (4)
- If $k = 5$, what would each test point be classified as? (2)
- What is a disadvantage of using a kNN on a constrained device such as an embedded system? (2)

Question 9: Neural Networks [12 marks]

Suppose you have a fully-connected, feedforward neural network with the weights initialised as shown in Figure 3. All of the activation functions for each neuron are Rectified Linear Units (ReLU). The input data has two features and the target variable has one feature.

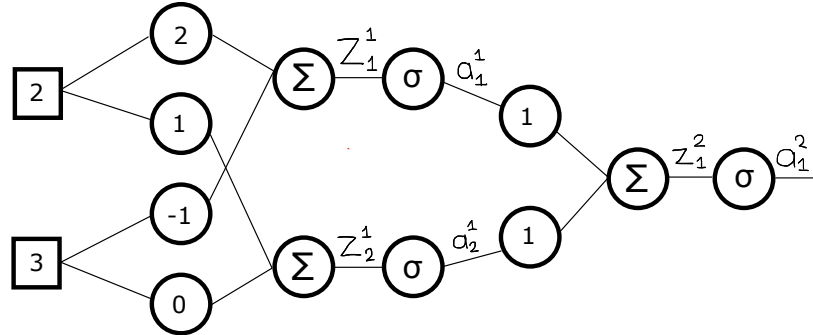


Figure 3: A fully-connected, feedforward neural network

The cost function you plan to optimize is given by: $J = \frac{1}{2}(\hat{y} - y^2)$

Where $\hat{y}$ is the predicted output of the model and y is the target variable.

The model is initialised with all bias values set to 1, but have been omitted from the diagram for simplicity purposes only.

a. Calculate the following values based on a single forward pass of the input.

- (i) $\mathbf{z}^1$, the net weighted inputs into all neurons in the first layer (1)
- (ii) $\mathbf{a}^1$, the activations of neurons in the first layer (1)
- (iii) $\mathbf{z}^2$, the net weighted inputs into all neurons in the second layer (1)
- (iv) $\mathbf{a}^2$, the activations of neurons in the second layer (1)

b. Using the chain rule, show that (4)

$$\frac{\partial J}{\partial w_{11}^1} = \delta^2 w_{11}^2 \sigma'(z_1^1) x_1$$

given that $\delta^l = \frac{\partial J}{\partial z^l}$

- c. Suppose the target variable is given by, $y = 2$ when $x = [2, 3]$. Using gradient descent calculate the updated value for w_{11}^1 given the learning rate, $\eta = 0.1$. (4)

Question 10: Reinforcement Learning [13 marks]

You have a mobile robot in the discrete grid world shown in Figure 4. There are five states (locations on the grid) and two possible actions it can take at each time step by moving West (W) or East (E) provided there is space to move in to. s_1 and s_5 are terminal states with reward values of +1 and +2 respectively. The non-terminal states have reward values as shown in Figure 4. The robot begins each trial in s_3 .

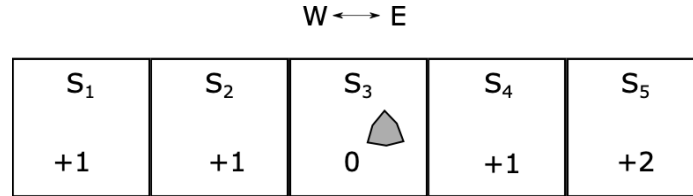


Figure 4: A mobile robot agent in a discrete grid world environment

You decide to use ϵ -greedy Q-learning to train a policy for the agent. All of the initial Q-values are set to zero. The Bellman equation below indicates how the Q-values are updated.

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R(s) + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

The learning rate is a fixed value given by $\alpha = 0.5$. The discount factor is set to $\gamma = 0.8$. Figure 5 shows the $N(s, a)$ (indicates number of times (s,a) is encountered) and $Q(s, a)$ ‘tables’ after the first trial. The relative positions of the values in each state cell indicate the action chosen e.g. The number left of the cell corresponds to ‘W’ and to the right is ‘E’. This is to make visualising the transitions easier.

$W \longleftrightarrow E$

0	0	0	0	0	1	0	1	1
---	---	---	---	---	---	---	---	---

$N(s, a)$

1	1	0	0	0	0	0	?	2	2
---	---	---	---	---	---	---	---	---	---

$Q(s, a)$

Figure 5: The $N(s, a)$ and $Q(s, a)$ tables after the first trial

- a. Calculate the value for $Q(s_4, E)$ after the first trial as indicated in Figure 5 (2)
- b. Suppose after the second trial the $N(s, a)$ table is given by Figure 6
 - (i) Write down the series of (state, reward, actions) taken that produced the $N(s, a)$ table. (1)
 - (ii) Draw the corresponding $Q(s, a)$ table (2)
- c. After the second trial would the agent have a preference to move W, E or have no preference? (2)
- d. Suppose two more trials are conducted such that the agent repeats the actions taken in trial 1 and then repeats the actions taken in trial 2.
 - (i) Calculate the $Q(s, a)$ values for $Q(s_3, W)$ and $Q(s_3, E)$ (4)

$$W \longleftrightarrow E$$

1	1	0	1	1	0	1	1
---	---	---	---	---	---	---	---

$$N(s, a)$$
Figure 6: The $N(s,a)$ table for t_2

- (ii) After the above trials would the agent have a preference to move W, E or no preference? (2)
Explain your answer.

END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x^*}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2 z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$